CORTEX — Systematic Literature Search (SLS)

Module Specification v1.1 — February 2026

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1. Overview

The Systematic Literature Search (SLS) is the foundational module of the CORTEX Clinical Affairs suite. It manages the entire lifecycle of a regulatory-grade literature search: query construction, execution across multiple databases, AI-assisted screening, full-text PDF retrieval, and finalization into a locked, auditable dataset that feeds the SOA, CER, and PMS modules.

Core constraint: Every decision (inclusion, exclusion, score) must be traceable and justified for Notified Body audit.

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2. Concepts & Data Model

2.1 Hierarchy

```
Project (device/product)
└─ SLS Session (one systematic search campaign)
   ├─ Search Queries (versioned, iterable)
   │  └─ Query Execution Records (per database, per query, per run)
   ├─ Articles (pooled results from all databases)
   │  └─ Article-Query Links (which query/database returned this article)
   ├─ Screening Decisions (audit log)
   └─ Final Locked Dataset
```

Article-Query Link Record

Field	Description
article_id	Parent article
query_id	Which query returned this article
database	Which database (PubMed, Scholar, Cochrane, etc.)
execution_id	Which query execution run
raw_rank	Position in original database results (if available)

An article can have multiple links (found by multiple queries and/or in multiple databases). This is the foundation for accurate PRISMA flowchart generation with per-query, per-database breakdown.

Query Execution Record

Field	Description
id	UUID
query_id	Parent query
database	Target database
executed_at	Timestamp
query_string_used	The database-specific syntax actually executed
result_count	Number of results returned
articles_imported	Number actually imported into the pool
status	success / partial / failed / cancelled

This record is the source of truth for the PRISMA "Identification" box counts.

2.2 Article Statuses

An article moves through a linear funnel:

Status	Meaning
pending	Returned by search, not yet scored or reviewed
scored	AI relevance score assigned
included	Marked for inclusion (human or bulk action)
excluded	Marked for exclusion with a logged reason
skipped	Deferred — needs further review later
full_text_review	Included at abstract level, under full-text review
final_included	Confirmed included after full-text review
final_excluded	Excluded after full-text review (reason logged)

2.3 SLS Session Statuses

Status	Description	Editable?
draft	Queries being built and tested	Yes
screening	Query validated, results imported, screening in progress	Yes (articles only)
full_text_review	Abstract screening done, full-text review in progress	Yes (articles only)
locked	Finalized. Read-only. Feeds downstream modules.	No

3. Detailed Workflow

PHASE 1 — Query Construction

3.0 Upstream Dependency: Clinical Evaluation Plan (CEP)

Every SLS session must be linked to a CEP (Clinical Evaluation Plan) at the project level. The CEP defines the overall clinical evaluation strategy, including which literature searches are needed, the appraisal criteria, and the update schedule. The SLS scope and parameters should be consistent with the CEP's search strategy section.

When creating an SLS session, the system displays the relevant CEP excerpt for reference.

3.1 Scope Definition

Before building queries, the user defines the search scope on the SLS session. Scope fields are dynamic — they adapt based on the SLS purpose, since different search types focus on different dimensions:

SLS Purpose	Primary Scope Fields	Secondary / Optional
soa_clinical	Pathology, Population, Clinical context	Device type (if relevant)
soa_device	Similar Device(s), Intended Use, Technology type	Pathology, Population
similar_device	Device name/type, Manufacturer, Regulatory class	Intended Use, Indications
pms_update	Time window, Pathology, Device	New risks, competitor updates
ad_hoc	Free-form scope description	Any relevant field

Available scope fields (shown/hidden dynamically):

- Pathology (free text + suggested tags)
- Intended Use (linked from project, editable)

- Population (e.g., "Adults ≥18, no pediatric")
- Device specificities (e.g., "CT Perfusion software, AI-based")
- Similar device(s) (name, manufacturer, regulatory reference)
- Technology / Algorithm type (e.g., "Deep learning, U-Net, perfusion mapping")
- Free-form scope notes (always available, for nuance)

The user selects the search purpose first, which determines which fields are shown prominently vs. collapsed. All fields remain accessible if the user wants to add context.

This scope definition is the reference used by the AI for relevance scoring later — the more precise the scope, the better the scoring quality.

3.2 Query Builder — Two Modes

Mode A — Visual Query Builder (Manual)

A structured UX for building boolean queries:

- The query is composed of Groups (rows)
- Each Group contains Keywords (terms/phrases)
- Within a group: terms are joined by OR (synonyms)
- Between groups: groups are joined by AND (intersection) or OR (union) — user chooses per group connector
- Each keyword can include wildcards (e.g., `stroke*`, `"ischemic stroke"` for exact phrase)

Example:

```
Group 1 [AND]: "CT perfusion" OR "CTP" OR "perfusion imaging"
Group 2 [AND]: "stroke" OR "ischemic stroke" OR "cerebral infarction"
Group 3 [AND]: "software" OR "artificial intelligence" OR "deep learning" OR "machine learning"
Group 4 [NOT]: "animal" OR "pediatric" OR "neonatal"
```

The builder generates a preview of the final boolean string in real time.

Mode B — Natural Language (AI-Assisted)

- User types a prompt describing what they want (e.g., "Find studies on AI-based CT perfusion analysis for stroke detection in adults, last 5 years, no animal studies")
- The configured LLM translates this into a structured boolean query (same format as Mode A)
- The generated query appears in the Visual Query Builder for review and editing
- User can iterate: tweak the prompt → regenerate → edit manually → hybrid approach

Both modes produce the same underlying query structure. The user can switch freely between modes.

3.3 Search Parameters

Configured alongside the query:

Parameter	Details
Databases	Multi-select: PubMed, Cochrane, Embase, Google Scholar, Scopus, Web of Science. Displayed with status (API available / manual export needed).
Date range	Start date – End date
Languages	Multi-select with shortcut: "All European languages" (EN, FR, DE, ES, IT, PT, NL, PL, SV, DA, NO, FI, EL, CS, RO, HU, BG, HR, SK, SL, LT, LV, ET, GA, MT). Individual selection also possible.
Article types	Multi-select: RCT, Meta-analysis, Systematic Review, Cohort Study, Case-Control, Cross-sectional, Case Report, Review, Guidelines, Technical Report, Conference Abstract.

3.4 Query Iteration Workflow

The user can:

- Run a query → see result count **per database AND per query** + total (no article details yet, just counts)
- Modify the query → adjust keywords, groups, parameters → re-run
- Duplicate a query → fork it to test a variant without losing the original
- Compare query variants side-by-side (result counts, overlap estimate)
- Version history — every run is saved: query string, parameters, database, date/time, result count

Each execution creates a **Query Execution Record** (see 2.1) that stores the exact count per query per database. These records are the source data for PRISMA Identification counts.

Once satisfied, the user validates the query → status moves from draft to screening → full article data is fetched and loaded.

3.5 Database-Specific Query Adaptation

The stored query structure is database-agnostic. At execution time, the system translates it into the syntax of each selected database:

- PubMed: MeSH terms + boolean syntax + filters via E-utilities API
- Cochrane: Cochrane-specific syntax
- Embase: Emtree terms + Embase syntax
- Others: Best-effort translation

The user can view and manually override the per-database translation before execution.

PHASE 2 — Results Display & Deduplication

3.6 Article Results View

Once the query is validated and executed, all results from all databases are pooled into a single table:

Table columns:

- Checkbox (for bulk selection)
- Status badge (pending / scored / included / excluded / skipped)
- AI Score (0-100, once scored)
- Title
- Authors (truncated, expandable)
- Journal
- Year
- Database source(s) — an article found in multiple databases shows all sources
- Query source(s) — which query(ies) returned this article
- Article type
- Language
- Abstract (expandable inline or side panel)

Table features:

- Sortable by any column
- Filterable by: status, score range, database, **query**, article type, language, year range
- Full-text search across title + abstract
- Total count displayed prominently (e.g., "847 articles from 3 databases, 4 queries")
- Pagination or virtual scroll for large result sets

3.7 Automatic Deduplication

On import, articles are deduplicated based on:

1. DOI match (exact)
2. PMID match (exact)
3. Title similarity (fuzzy, >95% match) + same first author + same year

Duplicates are merged: the record is kept once, with all source databases and source queries recorded in Article-Query Links. Duplicate detection is logged in the audit trail.

Deduplication statistics are tracked for PRISMA:

Metric	Description
total_raw_records	Sum of all Query Execution Record result counts
unique_articles	Count of deduplicated articles in the pool
duplicates_removed	total_raw_records − unique_articles
duplicates_cross_database	Articles found in 2+ databases
duplicates_cross_query	Articles found by 2+ queries within the same database

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PHASE 3 — AI Screening & Bulk Actions

3.8 AI Relevance Scoring

Trigger: User clicks "Score Articles" (can score all pending, or a selection).

Process (async with progress):

- For each article, the configured LLM receives: the SLS scope definition + article title + abstract + the exclusion code list (see 3.12)
- The LLM returns:
  - Relevance score (0-100): how well this article matches the defined scope
  - Reasoning (short text): why this score (stored for audit)
  - Suggested exclusion code (if score < threshold): mapped to the project's configured exclusion code list (e.g., SS for scope mismatch, POP for wrong population)
- Progress bar shows: "Scoring article 234 / 847..."
- Cancellable at any time. Already-scored articles keep their scores.

LLM configuration: The system supports a configurable LLM backend (Claude, GPT, open-source). Default is Claude Sonnet. Configurable at project or system level.

3.9 Bulk Actions (Based on Score)

After scoring, the user can perform bulk operations:

Action	Description
Bulk Exclude below threshold	"Exclude all articles with score < X" — user sets X (e.g., 20). Each excluded article gets the AI-suggested exclusion code automatically logged.
Bulk Include above threshold	"Include all articles with score > Y" — user sets Y (e.g., 75).
Bulk Skip range	"Skip all articles with score between X and Y" — for manual review later.

Each bulk action:

- Shows a preview: "This will exclude 312 articles. Top 5 examples: [...]"
- Requires confirmation
- Is logged as a single audit event with the rule applied
- Each individual article's exclusion code is recorded

3.10 Optional: Custom AI Filter (Prompt-Based)

An advanced feature for secondary filtering:

- User writes a natural language criterion (e.g., "Does this article specifically mention Siemens CT scanners?")
- Selects which articles to run it on: included | excluded | pending | skipped | all
- The LLM evaluates each selected article against this criterion → returns a temporary secondary score (0-100) + yes/no answer
- Results displayed as an additional sortable/filterable column
- User can then bulk-act on the secondary score
- This secondary score does not replace the primary relevance score — it's an overlay

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PHASE 4 — Individual Article Review

3.11 Article Detail View

Clicking an article opens a detail panel (side panel or full page):

- Full bibliographic data (title, authors, journal, DOI, PMID, year, language, type)
- Full abstract
- AI relevance score + reasoning
- Any secondary scores from custom filters

- Current status with full history (audit log for this article)
- Action buttons: Include | Exclude (with code picker + optional free text) | Skip | Revert
- Notes field (free text, for internal comments — not part of the formal exclusion reason)
- PDF attachment (upload / download / view) — for full-text review phase
- Tags (user-defined, for organization)
- **Source traceability** — list of all Query-Database combinations that returned this article

3.12 Exclusion Reason Codes

When excluding manually or via bulk AI action, the user selects from a configurable exclusion code list. Each code has a **short code** (for PRISMA display and compact reporting) and a **full label**.

Default Exclusion Code List:

Short Code	Full Label	Description
POP	Wrong Population	Study population does not match intended use (pediatric, animal, etc.)
PATH	Wrong Pathology / Indication	Different clinical condition than targeted
SD	Study Design Excluded	Study design excluded by scope (e.g., editorial, letter, opinion)
SS	Subject / Scope Mismatch	General scope mismatch — article topic does not align with search objectives
OUT	Outcome Not Relevant	Reported outcomes/endpoints do not match the clinical evaluation objectives
DEV	Wrong Device Type	Device studied is not relevant (wrong modality, wrong application)
MA	Meta-analysis / Review (double-counting risk)	Excluded to avoid double-counting when individual studies are already included
CS	Case Study / Case Series (excluded by scope)	Case-level evidence excluded per protocol when higher-level evidence is available
NE	Not in Selected Language(s)	Article not available in any of the selected languages
NF	Not Found / Full-text Unavailable	Full-text could not be retrieved for full-text review
DUP	Duplicate	Duplicate of another included article (not caught by automatic deduplication)
ANI	Animal / In Vitro / Phantom Only	Pre-clinical study only
TECH	Not Clinical (Technical / Engineering)	Technical paper without clinical data
RET	Retracted	Article has been retracted
DATA	Insufficient Data	Article does not contain sufficient quantitative data for extraction
OTH	Other	Free text required when selected

Configuration rules:

- Codes are configurable per project: the user can add, rename, hide, or reorder codes
- Each project can define custom codes beyond the defaults
- Short codes must be unique within a project, max 6 characters, uppercase alphanumeric

- When hiding a default code, existing articles with that code retain their exclusion reason
- AI-suggested exclusion reasons map to these codes: the scoring prompt includes the code list, and the AI returns the most appropriate code along with its reasoning

Display:

- In the article list table: the exclusion reason column shows the **short code** with full label on hover
- In the PRISMA flowchart: exclusion reasons are displayed as **short codes with counts** (e.g., `SS = 68, CS = 22, NF = 14, NE = 3, ...`)
- In the exported article list (XLSX): both short code and full label columns

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PHASE 5 — Full-Text Review & PDF Management

3.13 Full-Text Phase Transition

Once abstract-level screening is complete, the user transitions the session to `full_text_review` status. At this point:

- Only `included` articles carry forward
- The user can trigger bulk PDF download (async, with progress):
  - System attempts to retrieve PDFs via: DOI resolution (Unpaywall API for open access), PubMed Central, institutional proxy if configured
  - PDFs that can't be auto-retrieved are flagged for manual upload
  - Progress: "Retrieved 42/67 PDFs. 25 require manual upload."
- Each article shows a PDF status: `available` | `downloading` | `not_found` | `manual_needed`

3.14 Full-Text Screening

- User reviews full-text PDFs within the app (embedded PDF viewer or download)
- Can change status to `final_included` or `final_excluded` (with exclusion code — same code list as abstract screening)
- Reference mining (Phase 6) is available at this stage

PHASE 6 — Reference Mining

3.15 Bibliography Extraction

For any `final_included` article with a PDF:

- User triggers "Extract References" (per article or bulk)

- The LLM parses the PDF bibliography section → extracts structured references (title, authors, year, journal)
- Extracted references are matched against existing articles in the session (dedup)
- New references are displayed as "Mined References" with a flag
- User can add mined references to the screening pool → they enter as `pending` with `source_type = reference_mining`
- One level deep only (no recursive mining)

PHASE 7 — Finalization & Lock

3.16 Session Finalization

When the user is satisfied with the final article set:

Validation checks before locking:

- All included articles have been moved to `final_included` or `final_excluded`
- No articles remain in `pending` or `skipped` status (warning if so)
- All `final_excluded` articles have a reason code
- PRISMA counts are consistent (auto-verified — see below)

PRISMA Flowchart — Live Calculated View

The PRISMA flowchart is **not a static document filled manually**. It is a **computed view** derived from the article pool, Article-Query Links, screening decisions, and Query Execution Records. Every screening decision updates the flowchart in real time.

The flowchart is accessible at any time during screening (not just at finalization), giving the user a live dashboard of search progress.

Flowchart structure — Identification (per database, per query):

For each database, the flowchart shows the breakdown by query:

Records identified through PUBMED database searching

Query 1 n = [count from execution record]

Query 2 n = [count from execution record]

Query 3 n = [count from execution record]

Query 4 n = [count from execution record]

Total = [sum]

Records identified through SCHOLAR database searching

Query 1 n = [count]

Query 2 n = [count]

Query 3 n = [count]

Query 4 n = [count]

Total = [sum]

Records identified through COCHRANE database searching

Query 1 n = [count]

...

Total = [sum]

Records identified through FREE searches

Query 1 n = [count]

Total = [count]

Total records found (n = [grand total across all databases])

Flowchart structure — Deduplication & Screening:

```
Records after duplicates removed (n = [duplicates_removed from 3.7])

Records screened on basis of titles and abstracts (n = [unique_articles])

Records excluded with reasons (n = [total excluded at abstract level])
  SS  = [count]
  CS  = [count]
  NE  = [count]
  SD  = [count]
  MA  = [count]
  OUT = [count]
  POP = [count]
  NF  = [count]
  DEV = [count]
  ANI = [count]
  TECH = [count]
  ...
  [only codes with count > 0 are shown]
```

Flowchart structure — Full-Text & Inclusion:

```
Full-text articles assessed for eligibility (n = [entered full_text_review])

Full-text articles excluded with reasons (n = [final_excluded count])
  [same code breakdown, only codes with count > 0]

Articles included from reference mining (n = [source_type = reference_mining AND final_included])

Studies included (n = [final_included total])
```

Consistency validation:

The system auto-verifies that the arithmetic is correct:

- Total records found = sum of all Query Execution Record counts
- Total records found − Duplicates removed = Records screened
- Records screened − Excluded at abstract − Skipped = Full-text assessed (warning if Skipped > 0)
- Full-text assessed − Excluded at full-text + Reference mining additions = Studies included

If any equation fails, a red warning is shown with the specific discrepancy. This prevents the "numbers don't add up" problem.

Export formats:

- SVG / PNG / PDF — visual flowchart diagram matching PRISMA 2020 layout

- Editable source — the user can adjust label text and layout before export (e.g., add manual annotations), but the **counts are always computed and not manually editable** (this ensures accuracy)

3.17 Exports

From any state (but especially when locked):

Export	Format	Description
PRISMA flowchart	SVG / PNG / PDF	Visual diagram with per-query, per-database identification breakdown, exclusion reason codes with counts, and full-text phase. Live-calculated, always consistent.
Article list (full)	XLSX	All articles, statuses, scores, exclusion codes (short + full label), source queries, source databases
Included articles only	XLSX or DOCX	Formatted table for CER annex
Search strategy report	DOCX	Scope, queries (all versions), databases, dates, parameters, result counts per query per database, reproducibility statements
Exclusion reason summary	XLSX	Pivot table: exclusion code × screening phase (abstract / full-text) with counts and percentages
RIS / BibTeX export	RIS / BIB	For Zotero or other reference managers

3.18 Reproducibility Statement (MDCG/CEAR Compliance)

The Search Strategy Report export includes an auto-generated reproducibility statement per database, ensuring an independent reviewer can reproduce the search. Format:

> **Reproducibility Statement — PubMed** This search can be reproduced by executing the following query on the PubMed database (https://pubmed.ncbi.nlm.nih.gov/) using the Advanced Search Builder:
> Query: ("CT perfusion" OR "CTP") AND ("stroke" OR "ischemic stroke") AND ("software" OR "artificial intelligence") NOT ("animal" OR "pediatric")
> Filters applied: Date range: 2019/01/01 – 2025/12/31; Languages: English, French, German; Article types: Clinical Trial, Meta-Analysis, Systematic Review, Observational Study
> Execution date: 2026-02-10
> Result count at execution: 847 articles
> Note: Result count may vary if executed on a later date due to newly indexed articles.

This statement is auto-generated from the stored query, parameters, and execution metadata. One statement per database. Included as an annex to the Search Strategy Report.

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4. Async Task System

Any operation that processes multiple articles or calls external APIs is asynchronous:

Operation	Estimated Duration	Async?
Query count check	2-5s	No (inline)
Query execution (full fetch)	10-60s depending on DB + count	Yes
AI scoring (batch)	1-3s per article × N	Yes
Custom AI filter	1-3s per article × N	Yes
Bulk PDF download	5-30s per PDF × N	Yes
Reference mining (batch)	10-30s per PDF × N	Yes
PRISMA generation (export)	2-5s	No (inline)
PRISMA live view refresh	< 1s (computed from DB)	No (inline)

Async task UX:

- Task appears in a task panel (bottom bar or sidebar): task name, progress (N/total), elapsed time, estimated remaining
- Cancel button on every running task — stops processing, keeps already-completed items
- Tasks can run in background while user navigates other parts of the module
- On completion: toast notification + task panel updates
- Task history available (completed, cancelled, failed with error details)

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5. Access & Permissions

Role	Can Do
clinical / regulatory	Full access: create sessions, build queries, screen, lock
manager	Full access + can unlock a locked session (with audit log entry + justification)
collaborateur	Read-only access to locked sessions and exports
admin	Everything + system configuration (LLM settings, exclusion code lists, default templates)

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6. Database Integrations — Technical Details

6.1 API Connections

Database	Method	API	Cost	Rate Limit
PubMed	Direct API	NCBI E-utilities (esearch + efetch)	Free	10 req/s (with API key)
Cochrane	PubMed subset + manual fallback	cochranelibrary[sb] filter in E-utilities	Free	Same as PubMed
Google Scholar	SerpAPI	google_scholar engine	~€50/mo (5,000 searches)	5,000 searches/mo
Embase / Scopus	Manual import (RIS/CSV)	Elsevier API if institutional access available	Subscription	v1: import only
Web of Science	Manual import (RIS/CSV)	Clarivate API if institutional access available	Subscription	v1: import only

For databases without API access, the system provides a guided import flow: instructions for running the query on the database website → export as RIS/CSV → upload into CORTEX → automatic parsing and merge with the article pool. Each imported file is linked to a Query Execution Record for PRISMA traceability.

The query builder generates per-database syntax previews for manual execution on external platforms.

6.2 PDF Retrieval Strategy

Priority: free auto-retrieval first, then manual upload with smart verification.

Free auto-retrieval sources (in order):

1. PubMed Central (PMC) — via E-utilities efetch with `db=pmc`. Free full-text for all PMC-indexed articles (~8M+).
2. Unpaywall API — free, no key needed, 100K req/day. Pass a DOI → get open-access PDF URL if available (~30-40% of biomedical literature).
3. Europe PMC — additional European-funded open access content.
4. DOI resolution — follow DOI link, check for open-access landing page.

Manual upload with smart verification:

For PDFs not available from free sources:

1. Article is flagged with status `pdf_manual_needed`

2. User uploads a PDF file
3. System extracts text from first page / header via LLM
4. Automatic verification: extracted title + authors compared against the article record
5. Verification result displayed:
   - ✅ Match confirmed — title + authors match
   - ⚠️ Possible mismatch — shows extracted vs. expected, user confirms or rejects
   - ❌ Wrong article — clear mismatch, user must re-upload
6. If the uploaded PDF matches a different existing article in the pool, the system detects and flags it

Manual article addition (unknown articles):

Users can add articles not originating from any search (theses, internal reports, conference proceedings):

- Upload PDF → LLM extracts metadata (title, authors, year, journal/source)
- User reviews and edits the extracted metadata
- Article enters the pool with `source_type = 'manual_add'`
- Subject to the same screening funnel (scoring, include/exclude)

6.3 Reference Mining Pipeline

Target accuracy: ~90% via a hybrid extraction pipeline:

1. **GROBID** (self-hosted, open-source) — ML-based PDF parser specialized for scientific articles. Extracts structured bibliography with title, authors, year, journal, DOI. ~85% accuracy standalone.
2. **LLM fallback** — references GROBID can't parse cleanly are sent to the configured LLM for structured extraction from raw text.
3. **CrossRef + PubMed validation** — extracted references are matched against CrossRef `/works` API (free, ~50 req/s) and PubMed to find canonical records with DOI/PMID/abstract.
4. **Deduplication** — matched references checked against existing article pool.
5. **User review** — unmatched references (~10%) presented for manual handling.

Cost: free (GROBID self-hosted Docker, CrossRef API free polite pool).

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7. LLM Abstraction Layer

7.1 Supported Backends

Backend	Provider	Use Case	API Endpoint
Claude	Anthropic	Default. Best reasoning quality.	api.anthropic.com/v1/messages

Backend	Provider	Use Case	API Endpoint
ChatGPT	OpenAI	Alternative cloud LLM.	api.openai.com/v1/chat/completions
Ollama	Local	On-premise, zero-cost bulk operations, data-sensitive contexts.	localhost:11434/api/chat

7.2 Architecture

A unified abstraction layer normalizes all LLM calls:

```
callLLM({ provider, model, system, messages, temperature, maxTokens })
  → { content, usage, model, latency }
```

Each provider has an adapter handling API-specific translation. Adding new providers (Mistral, Gemini, etc.) requires only a new adapter module.

7.3 Configuration

- System-level default: set in CORTEX admin settings (provider + model)
- Project-level override: a project can use a different provider/model
- Per-task override: specific operations (e.g., scoring vs. extraction) can use different models for cost optimization (e.g., Ollama for bulk scoring, Claude for complex extraction)
- Ollama settings: endpoint URL (default `http://localhost:11434`), model name (e.g., `llama3`, `mistral`, `mixtral`)

7.4 Cost Optimization

For bulk operations (scoring 800 articles), the system can use a cheaper/local model. For high-stakes operations (extraction grids, digest generation, CER drafting), it defaults to the highest-quality model. This is configurable per operation type.

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8. Open Items

1. **Google Scholar pagination** — SerpAPI returns max 100 results per query. For large result sets, pagination handling or query refinement guidance needed.
2. **Embase API** — institutional access varies. v1 is import-only; API integration added when Elsevier access is confirmed.
3. **PRISMA 2020 vs. classic** — the flowchart follows PRISMA 2020 structure. Export template should support both PRISMA 2009 (classic) and PRISMA 2020 layouts, selectable by the user.
4. **Exclusion code mapping across projects** — when importing SLS sessions between projects, exclusion codes must be mapped if the lists differ. A mapping UI is needed for this edge case.

5. **Manual import deduplication** — for databases with manual RIS/CSV import, the system must handle the case where the same article appears in both an API-fetched database and a manually imported one. DOI/PMID matching is the primary key; title fuzzy matching is the fallback.